

In this lecture, we give several striking applications of Gale diagrams and oriented matroid duality, among them

- the construction of an 8-polytope with 12 vertices that is nonrational, meaning that it cannot be realized with rational vertex coordinates (due to Perles),
- the construction of a 5-polytope for which the shape of a 2-dimensional face cannot be prescribed (new!), and
- the construction of a 24-polytope with 28 vertices for which the realization space is not connected.

These examples are “easily” derived from special low-dimensional point configurations via Gale diagrams.

Finally, we develop the “Lawrence construction,” a systematic method by which properties of arbitrary point configurations can be encoded into polytopes. This makes it possible to provide the corresponding “universality theorems,” which say that the realization spaces of convex polytopes can be arbitrarily bad, in a sense that we make precise ahead.

6.1 Circuits and Cocircuits

For this section, let $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathbb{R}^d$ be a finite set of n points in affine space $\mathbb{R}^d$ — for example, the vertices of a d -polytope. There is no problem with multiple points, but we always assume that the points in X affinely span $\mathbb{R}^d$. We continue our habit to interpret X as a matrix $X \in \mathbb{R}^{d \times n}$ when this is convenient.

We now explore two “dual” ways to derive combinatorial data from such a point configuration.

(a) Affine Dependences

The *affine dependences* of the point configuration X are the vectors $\mathbf{z} \in \mathbb{R}^n$ with $\mathbb{1}\mathbf{z} = 0$ such that $X\mathbf{z} = \mathbf{0}$. These vectors $\mathbf{z}$ form a vector subspace of $\mathbb{R}^n$,

$$\text{a-Dep}(X) := \{\mathbf{z} \in \mathbb{R}^n : X\mathbf{z} = \mathbf{0}, \mathbb{1}\mathbf{z} = 0\}.$$

Geometrically, let $\mathbf{z} \neq \mathbf{0}$ be an affine dependence. Then we can look at the sets of negative and positive coefficients of $\mathbf{z}$, $N(\mathbf{z}) := \{i : z_i < 0\}$ and $P(\mathbf{z}) := \{i : z_i > 0\}$. We get that $\Lambda := \sum_{i \in P(\mathbf{z})} z_i = -\sum_{i \in N(\mathbf{z})} z_i > 0$. By multiplying with $\frac{1}{\Lambda}$ we can rewrite the affine dependence $\mathbf{z}$ as

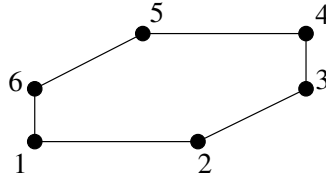
$$\sum_{i \in P(\mathbf{z})} \frac{z_i}{\Lambda} \mathbf{x}_i = \sum_{i \in N(\mathbf{z})} \frac{-z_i}{\Lambda} \mathbf{x}_i =: \mathbf{y},$$

so that

$$\mathbf{y} \in \operatorname{conv}\{\mathbf{x}_i : i \in P(\mathbf{z})\} \cap \operatorname{conv}\{\mathbf{x}_i : i \in N(\mathbf{z})\}$$

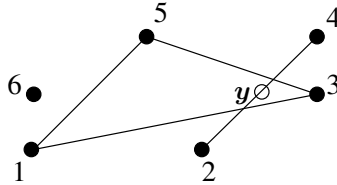
represents a point that lies both in the convex hull of the points with positive coefficients and in the convex hull of the points with negative coefficients.

For example, let $X = \begin{pmatrix} 0 & 3 & 5 & 5 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 1 \end{pmatrix}$ be the vertex set of a hexagon



Then the affine dependence

$$\mathbf{z} = \begin{pmatrix} 1 \\ -4 \\ 6 \\ -4 \\ 1 \\ 0 \end{pmatrix} \text{ represents } \mathbf{y} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \operatorname{conv}\{\mathbf{x}_1, \mathbf{x}_3, \mathbf{x}_5\} \cap \operatorname{conv}\{\mathbf{x}_2, \mathbf{x}_4\}.$$



In particular, we are interested in the special case of affine dependences involving a minimal set of points from X , that is, dependences of nonempty point sets such that every proper subset is affinely independent.

We define the *support* of a vector as the set of components that are not zero. Thus the minimal affinely dependent point sets correspond to the dependences with inclusion-minimal supports.

For the minimal dependences, we see that $\operatorname{conv}\{\mathbf{x}_i : i \in P(\mathbf{z})\}$ and $\operatorname{conv}\{\mathbf{x}_i : i \in N(\mathbf{z})\}$ are simplices whose relative interiors intersect in a unique point $\mathbf{y}$. Such configurations are known as “minimal Radon partitions” because of *Radon’s theorem*, a quite trivial but basic lemma from convexity theory; see Exercise 6.0.

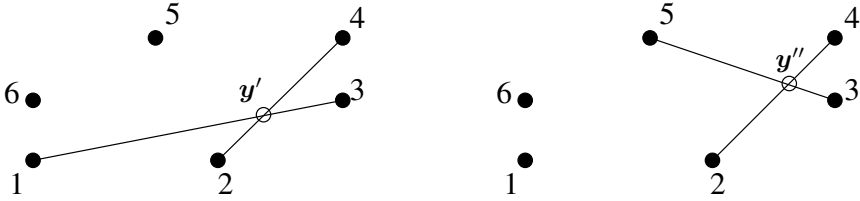
It is not hard to show that every affine dependence $\mathbf{z}$ is a finite sum of such minimal dependences, $\mathbf{z}' + \mathbf{z}'' + \dots + \mathbf{z}^{(k)}$. By Carathéodory’s Theorem 1.15 (see Lemma 6.7 below) the minimal dependences $\mathbf{z}^{(i)}$ can be chosen consistent with $\mathbf{z}$, that is, such that the j th component of each $\mathbf{z}^{(i)}$ either has the same sign as z_j or else it vanishes. Finally, the minimal affine dependences determine the point configuration X up to affine coordinate

change (Exercise 6.1). Together these three facts mean that the minimal affine dependences completely determine the structure of the point configuration.

For the hexagon we considered before, the linear dependence can be written as a sum

$$\begin{pmatrix} 1 \\ -4 \\ 6 \\ -4 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{5}{2} \\ 3 \\ -\frac{3}{2} \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{3}{2} \\ 3 \\ -\frac{5}{2} \\ 1 \\ 0 \end{pmatrix}$$

which writes it as a sum of the two minimal dependences depicted here:



To distill the “combinatorial essence,” we use the *sign function*

$$\text{sign} : \mathbb{R} \longrightarrow \{+, -, 0\}, \quad z \longmapsto \text{sign}(z) = \begin{cases} + & \text{if } z > 0, \\ 0 & \text{if } z = 0, \\ - & \text{if } z < 0. \end{cases}$$

We will apply the sign function componentwise to vectors, so for a column vector $\mathbf{z} \in \mathbb{R}^n$ we get a column sign vector $\text{sign}(\mathbf{z}) \in \{+, -, 0\}^n$, and for a row vector $\mathbf{c} \in (\mathbb{R}^n)^*$ we get a row sign vector $\text{sign}(\mathbf{c})$.

Definition 6.1. Let $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathbb{R}^d$ be a set of n points in affine space $\mathbb{R}^d$.

The *signed vectors* of X are the column sign vectors $\text{sign}(\mathbf{z})$ corresponding to affine dependences of the points in X ,

$$\mathcal{V}(X) := \{\text{sign}(\mathbf{z}) : \mathbf{z} \in \mathbb{R}^n, X\mathbf{z} = \mathbf{0}, \mathbf{1}\mathbf{z} = 0\} = \text{SIGN}(\text{a-Dep}(X)),$$

where $\text{SIGN}(U)$ denotes $\{\text{sign}(\mathbf{x}) : \mathbf{x} \in U\}$ for any subset $U \subseteq \mathbb{R}^n$.

The *signed circuits* of X are the column sign vectors $\text{sign}(\mathbf{x})$ corresponding to minimal affine dependences of the points in X . The set of signed circuits of X is denoted by $\mathcal{C} = \mathcal{C}(X)$.

For example, the 6-point configuration discussed above has

$$\text{the vector } \begin{pmatrix} + \\ - \\ + \\ - \\ + \\ 0 \end{pmatrix}, \quad \text{and the circuits } \begin{pmatrix} + \\ - \\ + \\ - \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ - \\ + \\ - \\ + \\ 0 \end{pmatrix},$$

corresponding to the nonminimal dependence (“the segment $[2, 4]$ intersects the triangle $[1, 3, 5]$ ”) and to the two minimal dependences (“the segment $[2, 4]$ intersects the segment $[1, 3]$, respectively the segment $[3, 5]$ ”) that we have calculated above.

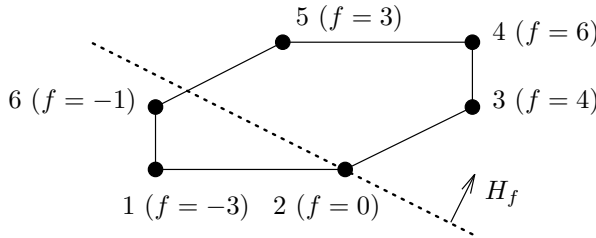
The sets $\mathcal{C}(X)$ and $\mathcal{V}(X)$ of signed circuits and vectors thus are combinatorial data associated with any affine point configuration. We will now proceed to extract the natural “dual data,” called signed cocircuits and signed covectors. In Section 6.2 we will derive the theoretical framework that shows that these four types of data are all “equivalent,” and that makes the “duality” precise.

(b) Affine Functions

From every affine function f on $\mathbb{R}^d$, of the form $\mathbf{x} \mapsto f(\mathbf{x}) = \mathbf{c}\mathbf{x} - z$, for $\mathbf{c} \in (\mathbb{R}^d)^*$, $z \in \mathbb{R}$, we get a row vector, called an *affine value vector*,

$$(f(\mathbf{x}_1), \dots, f(\mathbf{x}_n)) = (\mathbf{c}\mathbf{x}_1 - z, \dots, \mathbf{c}\mathbf{x}_n - z) = \mathbf{c}X - z\mathbb{1},$$

which records the values of $\mathbf{c}\mathbf{x} - z$ on the points $\mathbf{x}_i$ of X . For the hexagon from our example, the affine function $f(\mathbf{x}) = x_1 + 2x_2 - 3$ generates the affine value vector $(-3, 0, 4, 6, 3, -1)$:



where our drawing lists the values of f in brackets, and illustrates the hyperplane $H_f := \{\mathbf{x} \in \mathbb{R}^d : f(\mathbf{x}) = 0\}$ as a dashed line.

We note that the set

$$\text{a-Val}(X) := \{\mathbf{c}X - z\mathbb{1} \in (\mathbb{R}^n)^* : \mathbf{c} \in (\mathbb{R}^d)^*, z \in \mathbb{R}\}$$

of affine value vectors is a vector subspace of $(\mathbb{R}^n)^*$.

Geometrically, the row vector $f(X) := \mathbf{c}X - z\mathbb{1}$ records “signed distances” of the points $\mathbf{x}_i$ from the oriented hyperplane H_f . Thus the row sign vector $\text{sign}(f(X))$ records which points of X are on the positive side $H_f^+ \setminus H_f = \{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) > 0\}$ of the hyperplane H_f , on H_f itself, or on the negative side of H_f .

For example, for the hexagon and the affine function illustrated above, we have $f(X) = (-3, 0, 4, 6, 3, -1)$, and thus $\text{sign}(f(X)) = (-, 0, +, +, +, -)$.